From Deep UV to Mid-IR
OPTO DIODE
An ITW Company

Silicon Avalanche Photodiode Φ500um

ODD-APD-002

FEATURES

- 500um Circular Active Area
- Fast Response
- Low Capacitance
- Hermetically Sealed Package
- Low Noise
- High Gain

APPLICATIONS

- Time of Flight (ToF) LiDAR Systems
- Laser Rangefinders
- Autonomous Vehicle Obstacle Detection
- Safety Scanners
- Free-Space Optical Communications
- Optical Speed Measurements

Electro-Optical Characteristics at 25 °C

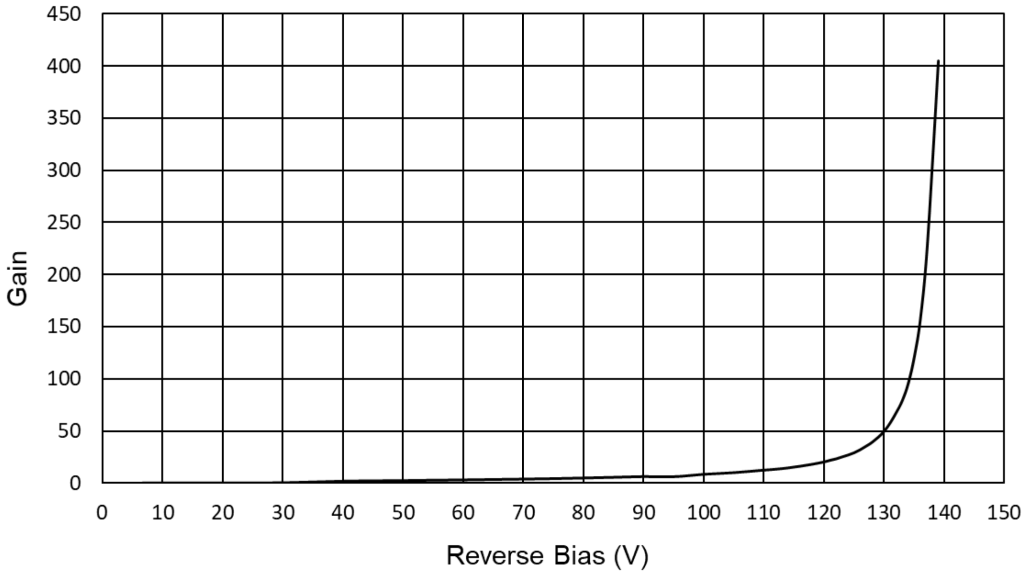
Parameters	Test Conditions	Min	Typ.	Max	Units
Active Area	Φ 0.5 mm		0.2		mm^2
Spectrum Response	-	400 ~ 1100			nm
Responsivity	λ = 905nm, M = 1	0.5	0.55	-	A/W
V_{OP} (Gain = M)	λ = 905nm, M = 100	95	-	209	V
	λ = 905nm, M = 30	34	-	50	
Dark Current, I_d	M = 100	0.2	0.4	1.0	nA
Rise Time, t_s	λ = 905nm, f = 1MHz, R_L = 50Ω,	-	0.5	1.5	ns
Capacitance, C	M = 100, f = 1MHz	-	1.2	-	pF
Temperature Coefficient	I_R = 10uA, -40°C ~ 85°C	-	0.9	-	V/°C
Reverse Breakdown Voltage	I_R = 10μA	100	-	220	V
Package	-	TO - 46			-

Thermal Parameters

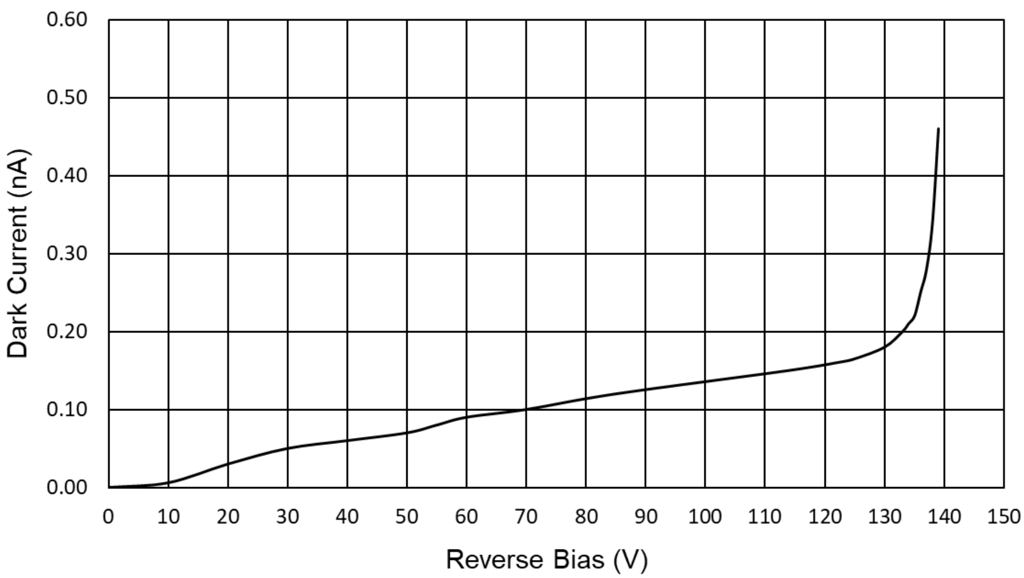
PARAMETER	MIN	MAX	Units
APD Supply Voltage	-	0.95*V_{BR}	V
Operating Temperature	-45	+85	°C
Storage Temperature	-45	+100	°C
Forward Current, I_F	-	5	mA
Lead Soldering Temperature[1]	-	260	°C

[1] 0.080" from case for 10 seconds.

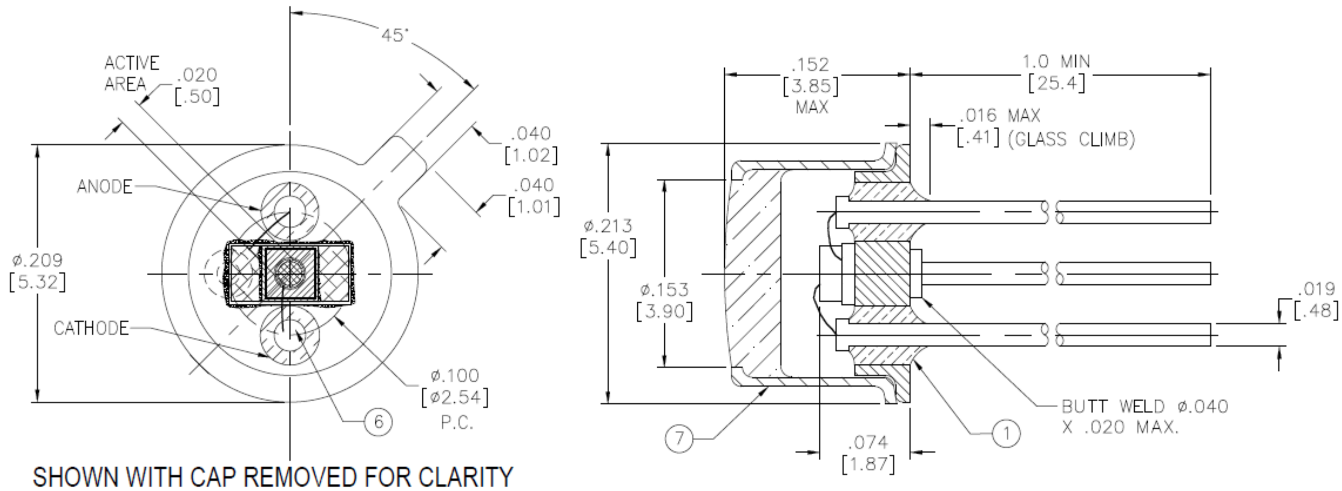
Typical Gain vs. Reverse Bias



Typical Dark Current vs. Reverse Bias



Package Information



Dimensions are in inch [metric] units.

Specifications are subject to change without prior notice.